

CS313 ADVANCED PROGRAMMING LANGUAGE

Car Rental System

A full-stack JavaFX application with 11 UI screens, 13 controllers, and enterprise architecture design patterns

2024

Complete platform with visual interface showcase and technical implementation details

What It Is

A complete rental management platform

Car Rental System transforms the complexity of multi-role rental operations into an elegant, user-friendly platform. It features 11 dedicated screens, role-based access control, and real-time inventory management.

AUTHENTICATION LAYER

Secure login with role-based routing to customer, agent, or admin dashboards

PROCESSING ENGINE

CRUD operations, booking validation, real-time status management

USER INTERFACES

11 FXML screens covering all functional workflows and user types

DATA MANAGEMENT

Complete inventory tracking, booking history, user management

CUSTOMER VIEW

IDS Car Rentals

Login & Join

Please enter your information

Customer enters
password
EX: customer@cc

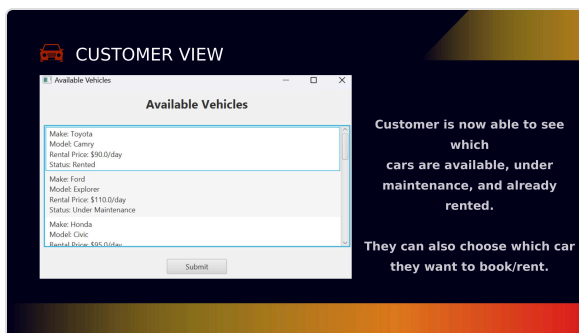
Car Agent enters
password
EX: agent@ag

Admin enters
password
EX: admin@ad

Login Screen - Secure authentication portal

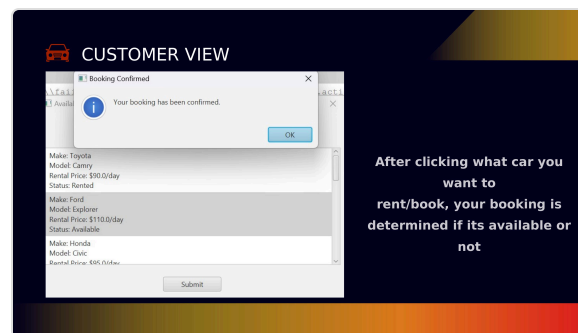
Customer Interface Screens

4 screens for customers to browse, book, and manage rentals



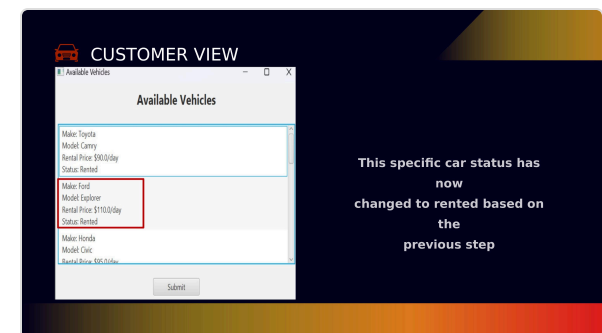
Browse Vehicles

Complete vehicle inventory display showing real-time availability status, pricing, and rental details



Create Booking

Customer selects desired vehicle and confirms booking with system validation

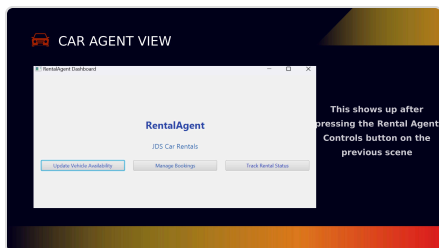


Booking Confirmed

Confirmation screen with automatic status update to Rented across system

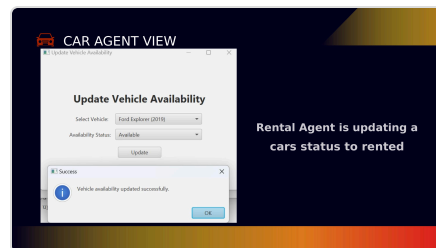
Rental Agent Interface Screens

4 screens for agents to manage operations and track active rentals



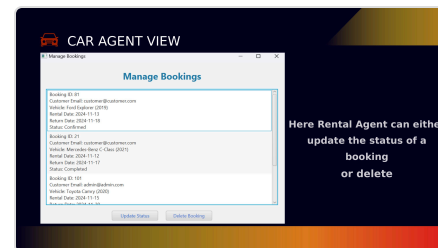
Dashboard

Operations hub for agents



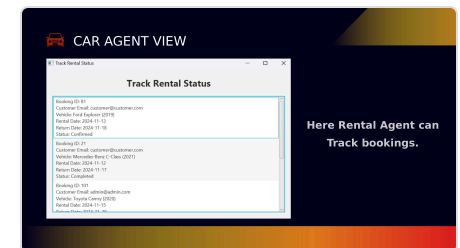
Update Status

Change vehicle status



Manage Bookings

Active booking control

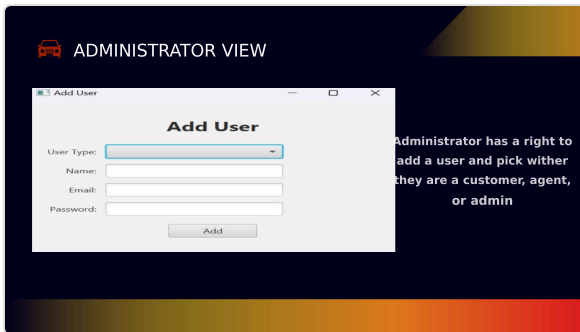


Track Rentals

Real-time monitoring

Administrator Interface Screens

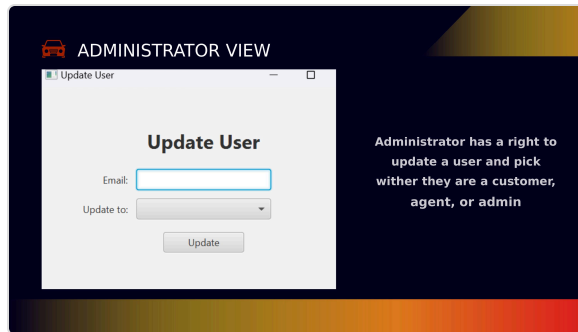
3 screens for system administration and user management



The screenshot shows the 'ADMINISTRATOR VIEW' window with a title bar icon of a bus. A modal window titled 'Add User' is open. It contains a 'User Type' dropdown menu, and three text input fields for 'Name', 'Email', and 'Password'. An 'Add' button is at the bottom. To the right of the modal, text states: 'Administrator has a right to add a user and pick whether they are a customer, agent, or admin'.

Add User

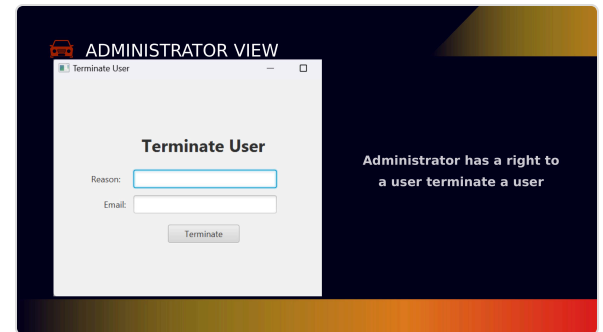
Create new accounts with role assignment



The screenshot shows the 'ADMINISTRATOR VIEW' window. A modal window titled 'Update User' is open. It contains a text input field for 'Email' and a dropdown menu for 'Update to:'. An 'Update' button is at the bottom. To the right of the modal, text states: 'Administrator has a right to update a user and pick whether they are a customer, agent, or admin'.

Update User

Modify user information and roles



The screenshot shows the 'ADMINISTRATOR VIEW' window. A modal window titled 'Terminate User' is open. It contains a text input field for 'Reason' and a text input field for 'Email'. A 'Terminate' button is at the bottom. To the right of the modal, text states: 'Administrator has a right to a user terminate a user'.

Terminate User

Deactivate or remove accounts

Technical Foundation

22

JAVA FILES

13

CONTROLLERS

11

UI SCREENS

7

MODEL CLASSES

Layered Architecture

View Layer

11 FXML screens, JavaFX UI components, Event binding, Real-time updates

Controller Layer

13 Controllers, Event handlers, Business logic, Form validation, Role-based routing

Model Layer

7 Domain classes, Abstract inheritance, Interface contracts, Data encapsulation

Database Layer

SQL queries, JDBC connections, Transaction management, Data persistence

Core Features and Capabilities

A

Authentication

Secure login with role-based routing

V

Vehicle Management

Real-time inventory tracking

B

Booking System

Complete rental workflow

U

User Management

Full CRUD operations

A

Agent Controls

Booking and status management

A

Admin Panel

System administration suite

Design Excellence: The system demonstrates clean code principles, SOLID design patterns, proper separation of concerns, and enterprise-grade architecture practices making it exemplary of professional software engineering

Implementation Quality

Code Organization

Clean separation, single responsibility

OOP Principles

Encapsulation, Inheritance, Polymorphism

Data Model

7 Domain classes with inheritance

Performance

Efficient queries, real-time updates

Controllers

13 specialized handlers

Patterns

MVC, Singleton, Factory, Observer

13 Specialized Controllers

LoginController, MainController, UserController, VehiclesController, AdminController, AddUserController, UpdateUserController, TerminateUserController, RentalAgentController, ManageBookingsController, TrackRentalStatusController, BookingController, UpdateVehicleAvailabilityController

Car Rental System

CS313 Advanced Programming Language

2024

A comprehensive demonstration of enterprise software architecture, clean code design patterns, and professional Java development practices.

Complete with 11 user interfaces, 13 controllers, 7 model classes, and role-based access control across a full-stack application.

All interfaces showcase real-world usage patterns and professional design.